

NATIONAL INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

B.Tech CSE, Semester 8

PROGRAMMING LAB · INTERNAL ASSESSMENT

SET 1

Time allowed: 1 hour

Maximum Marks: 25

Answer in either C or Python. Write each program in full.

The marks for questions are shown in brackets.

SECTION A

(Attempt all questions from this Section)

1. Write a program that reads a sorted array of n integers and a target value, and prints the element of the array closest to the target. [8]
2. Answer the following about the program you wrote for question 1:
 - (i) State the time complexity of your solution. [2]
 - (ii) Show a dry run for the array 3, 4, 5, 6, 9 with the target 5.2. [3]

SECTION B

(Attempt all questions from this Section)

3. Write a program that reads an array of 0s and 1s and prints the length of the longest run of 1s in that array.
 - (a) Reject the input and print a message if any element is not 0 or 1. [3]
 - (b) Print the maximum run length that was found. [4]

SECTION C

(Attempt any one question from this Section)

4. Rewrite your answer to question 3 so that it makes only one pass over the array. [5]
5. Explain why a sorted array allows question 1 to be solved without examining every element. [5]